Practical 11 - Code and Output

Yacc file: prac11.y

prac11.y

```
%{
#include <stdio.h>
#include <string.h>
int yylex(void);
void yyerror(char *);
int temp = 1;
%}

%union { char str[32]; }
%token <str> NUMBER
%type <str> expr
%left '+' '-'
%left '*' '/'

%%
input: | input stmt;
stmt: { temp = 1; } expr '\n';
expr:  NUMBER { strcpy($$, $1); }
     | expr '+' expr { printf("\t%d = %s + %s\n", temp, $1, $3); sprintf($$, "%d", temp++); }
     | expr '-' expr { printf("\t%d = %s - %s\n", temp, $1, $3); sprintf($$, "%d", temp++); }
     | expr '*' expr { printf("\t%d = %s * %s\n", temp, $1, $3); sprintf($$, "%d", temp++); }
     | expr '/' expr { printf("\t%d = %s / %s\n", temp, $1, $3); sprintf($$, "%d", temp++); }
     | '(' expr ')' { strcpy($$, $2); }
     ;
%%
void yyerror(char *s) { printf("Error: %s\n", s); }
int main() { return yyparse(); }

Name:- Mohammed Zaid Shaikh
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Batch:- A2
```

Lex file: prac11.l

prac11.l

```
%{
#include <stdio.h>
#include <string.h>
#include "y.tab.h"
%}

%%
[0-9]+   { strcpy(yyval.str, yytext); return NUMBER; }
[+-*/()-] { return yytext[0]; }
[ \t]    ;
\n       { return '\n'; }
.        { printf("Invalid character: %s\n", yytext); return yytext[0];  }
%%

int yywrap(void) { return 1;  }

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//Batch :- A2
```

Output

```
zaid787@LAPTOP-ID0Q783M:~$ gedit prac11.y
zaid787@LAPTOP-ID0Q783M:~$ gedit prac11.l
zaid787@LAPTOP-ID0Q783M:~$ yacc -d prac11.y
zaid787@LAPTOP-ID0Q783M:~$ flex prac11.l
zaid787@LAPTOP-ID0Q783M:~$ gcc y.tab.c lex.yy.c -o prac11
zaid787@LAPTOP-ID0Q783M:~$ ./prac11
2+3*4
t1 = 3 * 4
t2 = 2 + t1
(12+8)*(15-5)/2)+7*3-18/3
t1 = 12 + 8
t2 = 15 - 5
t3 = t2 / 2
t4 = t1 * t3
t5 = 7 * 3
t6 = t4 + t5
t7 = 18 / 3
t8 = t6 - t7
[1]+  Stopped                 ./prac11
zaid787@LAPTOP-ID0Q783M:~$
```